CS-3105 Machine Learning Lab

Lab 02 Report

NumPy for Numerical Computing

Field	Details
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1. Objective

The aim of this lab was to use NumPy to create, reshape, slice, filter and summarise arrays, and to apply those skills to four real datasets: Census 2023, climate normals for five cities, CPI inflation for FY2024 and FY2025, and a small set of class marks. The lab rules were followed throughout: no Python loops where a NumPy function exists, every output labelled with its task ID, and random results made reproducible with fixed seeds.

2. Tools and Environment

- Python 3 with NumPy 2.4.4 in a Jupyter notebook (Restart and run all, no errors).
- Matplotlib for the optional image display in Task 10.
- Data: PBS Census 2023 and 2017, PMD/WMO 1991–2020 climate normals, Pakistan Economic Survey 2024-25 (CPI), and sample lab marks. The starter notebook's data cell was left unchanged.

3. Method and Results by Task

Tasks 1–2: Setup and class register

NumPy 2.4.4 was imported as `np`. The register arrays were made with `np.zeros(10)`, `np.ones(10)` and `np.full(10, 5)`. Roll numbers 10 to 50 came from `np.arange(10, 51)` (41 values) because the stop value is excluded. Even roll numbers used `np.arange(10, 51, 2)` (21 values). The seat map was `np.arange(9).reshape(3, 3)` and the identity matrix was `np.eye(3)`.

Task 3: Randomness and reproducibility

With `np.random.seed(2026)` the single uniform value was **0.219346**, followed by 25 standard-normal values (sample mean −0.0582, sample std 1.0307). After `np.random.seed(42)`, 10,000 dice were rolled with `np.random.randint(1, 7, size=10000)`.

Face	1	2	3	4	5	6
Count	1665	1692	1625	1672	1689	1657

The estimated P(six) was **0.1657** against the theoretical 1/6 = 0.1667, a gap of only 0.0010. Double sixes were counted without a loop using the boolean pair `is_six[:-1] & is_six[1:]`, giving **265**

occurrences (theory predicts about 278). Fixing the seed means anyone who reruns the notebook gets exactly these numbers.

Task 4: Spacing and reshaping

The 10 × 10 grid from 0.01 to 1.00 was built as `np.arange(1, 101).reshape(10, 10) / 100`, which avoids floating-point drift from a fractional step. `np.linspace(0, 1, 20)` gave 20 evenly spaced points with both ends included. Lahore's rainfall `rain[0]` was reshaped to (4, 3) and summed with `axis=1`.

Quarter	Q1 (Jan–Mar)	Q2 (Apr–Jun)	Q3 (Jul–Sep)	Q4 (Oct–Dec)
Rain (mm)	84.9	111.0	454.3	24.0

The wettest quarter has index 2 (Q3, the monsoon) with 454.3 mm.

Task 5: Exam hall seating plan

With `hall = np.arange(1, 26).reshape(5, 5)`:

Expression	Result	Seats the invigilator points at
`hall[2:, 1:]`	3 × 4 block	Rows 3–5, columns 2–5: seats 12–15, 17–20, 22–25
`hall[3, 4]`	20	One seat: row 4, column 5
`hall[0:3, 1:2]`	[[2], [7], [12]]	Column 2 of the first three rows
`hall[4]`	[21 22 23 24 25]	The whole last row
`hall[3:]`	2 × 5 block	Last two rows: seats 16–25

The sum of all seat numbers is **325** and the standard deviation is **7.2111**.

Task 6: Census 2023

Total population in 2023 was **247,541,947**. Growth was computed with `((pop_2023 / pop_2017) ** (1/6) - 1) * 100`.

Unit	Share of total (%)	Annual growth 2017–2023 (%)
Punjab	51.58	2.52
Sindh	22.50	2.56
Khyber Pakhtunkhwa	16.50	2.37
Balochistan	6.02	3.19
Islamabad	0.95	2.80
AJK	1.75	1.15
Gilgit-Baltistan	0.69	2.28

The most populous unit is Punjab. Units above 20 million people (mask `pop_2023 > 20_000_000`) are Punjab, Sindh and Khyber Pakhtunkhwa. The fastest-growing unit is Balochistan at 3.19 % per year.

> **Interpretation:** Every unit grew, between 1.15 % and 3.19 % per year. Balochistan grew fastest, while AJK grew slowest at 1.15 %, less than half the 2.62 % average of the other six units, so its share of the national population is slipping.

Task 7: Climate of five cities

`tmax` and `rain` both have shape (5, 12). Per-city results use `axis=1`; per-month results use `axis=0`.

City	Annual mean max (°C)	Hottest month	Months > 35 °C	Monsoon rain (mm)	Yearly rain (mm)	Monsoon share (%)
Lahore	30.48	May (38.9)	3	454.3	674.2	67.4
Karachi	32.51	May (35.7)	3	234.8	318.0	73.8
Islamabad	28.91	Jun (38.3)	3	763.0	1261.2	60.5
Peshawar	29.54	Jun (39.6)	4	172.3	541.3	31.8
Quetta	25.20	Jul (36.8)	3	17.9	300.2	6.0

The 5-city monthly average peaks in June at 37.60 °C and is lowest in January at 18.46 °C. Lahore's temperatures in °F range from 65.1 (January) to 102.0 (May and June). Lahore's May and June maxima tie at 38.9 °C and `argmax` returns the first, so May is reported. The wettest monsoon city is Islamabad with 763.0 mm, obtained by slicing `rain[:, 6:9]`.

> **Interpretation:** Karachi has the highest yearly mean maximum (32.51 °C) yet the lowest peak of all five cities (35.7 °C), which surprised me; the sea keeps it warm all year without extreme summer heat. Quetta was the other surprise: only 6.0 % of its rain falls in the monsoon, since its rain arrives in winter.

Task 8: Inflation FY2024 vs FY2025

Average inflation was **26.22 %** in FY2024 and **4.87 %** in FY2025. FY2024 months above 25 % (a mask on `cpi[0]`) were Jul, Aug, Sep, Oct, Nov, Dec and Jan. The monthly drop `cpi[0] - cpi[1]` in percentage points:

Month	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Drop	17.2	17.8	24.5	19.6	24.3	25.6	25.9	21.6	20.0	17.0

The biggest drop was in **January** (25.9 points, from 28.3 % to 2.4 %). Using nested `np.where`, the FY2025 labels were: Jul High; Aug, Sep, Oct Moderate; Nov to Apr Low.

> **Interpretation:** Inflation fell by 17 to 26 percentage points in every month. Prices were still rising, but far more slowly: from November onwards inflation was in the "Low" band and reached 0.3 % in April.

Task 9: Grace marks

`np.clip(marks + 5, 0, 100)` turned [45 78 92 38 66 97 49 55 71 83] into [50 83 97 43 71 100 54 60 76 88]. The pass rate (≥ 50) rose from 70.0 % to **90.0 %**. Two students passed only because of grace marks: indices **0 and 6** (marks 45 and 49), found with `(marks < 50) & (after >= 50)`. The z-scores are [−1.17, 0.57, 1.31, −1.54, −0.06, 1.47, −0.96, −0.64, 0.20, 0.83]; a check confirms mean 0.00 and std 1.00.

Task 10: An image is an array

The "7" was drawn on an 8 × 8 zero canvas with `img[1, 1:7] = 255`, `img[2:7, 5] = 255` and `img[4, 3:7] = 128`; the last assignment overwrites pixel [4, 5] with grey. Operations: flip with `img[:, ::-1]`, invert with `255 - img`, crop with `img[1:5, 3:7]` (4 × 4), and brighten with `np.clip(img + 80, 0, 255)`. The image has shape (8, 8), dtype int64 and **10** white pixels (6 in the top bar, 4 in the stem).

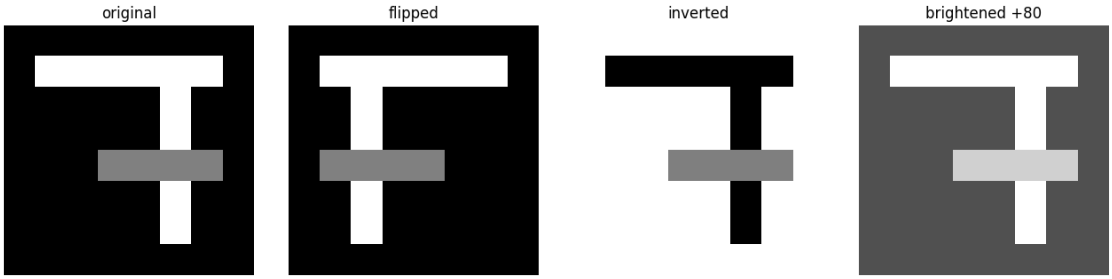


Figure 1. The 8 × 8 image: original, flipped, inverted and brightened.

4. Reflection

From the census I learnt that Balochistan grew fastest (3.19 % a year) while AJK grew slowest (1.15 %). From the climate data, Karachi has the highest yearly mean maximum (32.51 °C) but the lowest hottest-month value (35.7 °C), and only 6 % of Quetta's rain falls in the monsoon. From inflation, the average fell from 26.22 % to 4.87 % and every FY2025 month from November was "Low". `np.unique(..., return_counts=True)` saved me a loop: without it I would have written a `for` loop over 10,000 rolls, with six counters or a dictionary to update on each roll.

5. Rubric Self-Check

Criterion	How the notebook meets it
C1 Correctness of results	All 52 parts attempted; every value is computed from the data, never typed by hand.
C2 Array manipulation	`axis=1` per city, `axis=0` per month, `rain[:, 6:9]` for the monsoon, masks for > 20 M and > 35 °C, exact exam-hall slices.
C3 NumPy as a modern tool	No loops for computation; `reshape`, `clip`, `where`, `argmax`, `unique`, `sum(axis)`; seeds 2026 and 42 set.
C4 Interpretation and quality	Every output labelled with task ID and units; interpretations for Tasks 6, 7, 8 and 11; runs top to bottom without errors.

6. Conclusion

Vectorised NumPy replaced every loop in this lab with a single expression, and the same few ideas (the `axis` argument, boolean masks, slicing, broadcasting and seeded randomness) handled census, climate, inflation, marks and image data alike. These tools carry directly into Pandas in Lab 03 and the model-building labs that follow.